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Team 37

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Sentence Builder ER Diagram

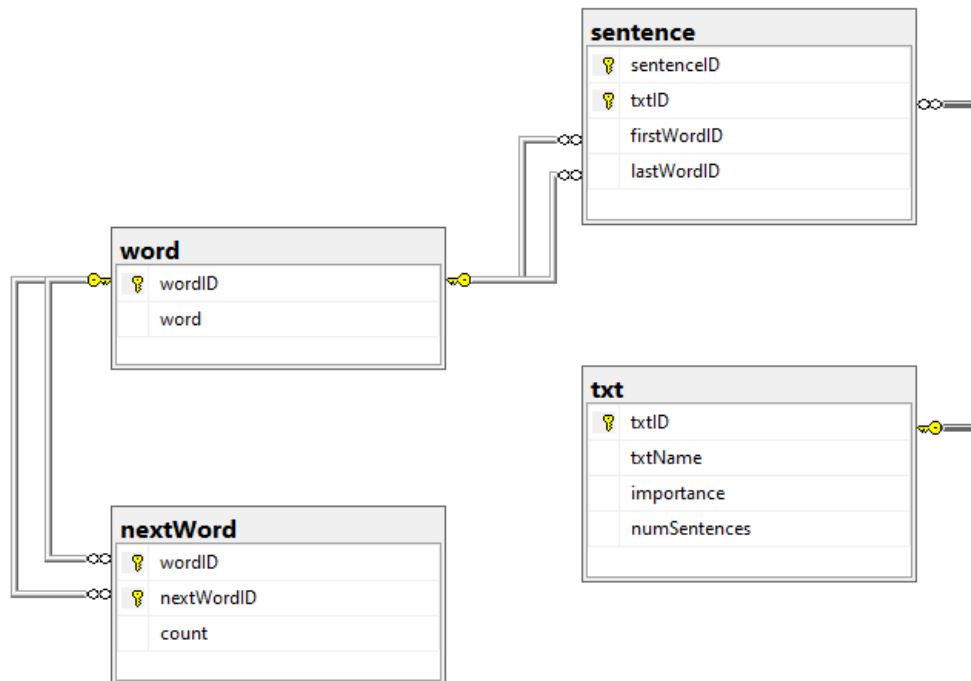


Figure 1. Entity-Relationship Diagram for the SentenceBuilder Database

Designed by Cortland Kimzey

Design Description

This database is used to store sentences and word relationships extracted from a txt file. The most important table in the database is the word table; this is made to store one instance of every word found in the txt files. This allows us to track what word comes first and last in a sentence in the sentence table as well as a count of the number of times a unique two-word combination appears in the nextWord table. We also have the txt table to track different information about our txt files.

Table Descriptions

word: A table to store a unique instance of every word found the in the txt files

wordID (Primary Key): Unique ID for each new word found

word: The actual word string

sentence: A table to store a unique instance of every sentence found tracking the first and last word in the sentence

sentenceID (Primary Key): Unique ID for each sentence found

txtID (Primary Key): Unique ID of the txt file the sentence came from

firstWordID (Foreign Key): The word ID of the first word in the sentence

lastWordID (Foreign Key): The word ID of the last word in the sentence

nextWord: A table to store every unique combination of two words and count the number of times it appears

wordID (Primary Key): Unique ID of the first word in the combo

nextWordID (Primary Key): Unique ID of the second word in the combo

count: The number of times the second word has been seen after the first

txt: A table to store information about each txt file used

txtID (Primary Key): Unique ID of a txt file used for data

txtName: name of the txt file to reference later if needed

importance: Rank of the txt file for data collection

numSentences: Number of sentences in the txt file